

ANSYS Practice Problems

Problem 1. Solve the 2-D heat conduction problem for the temperature distribution within the rectangular plate. Thermal conductivity of the plate, $K_{XX}=401 \text{ W/(m-K)}$.

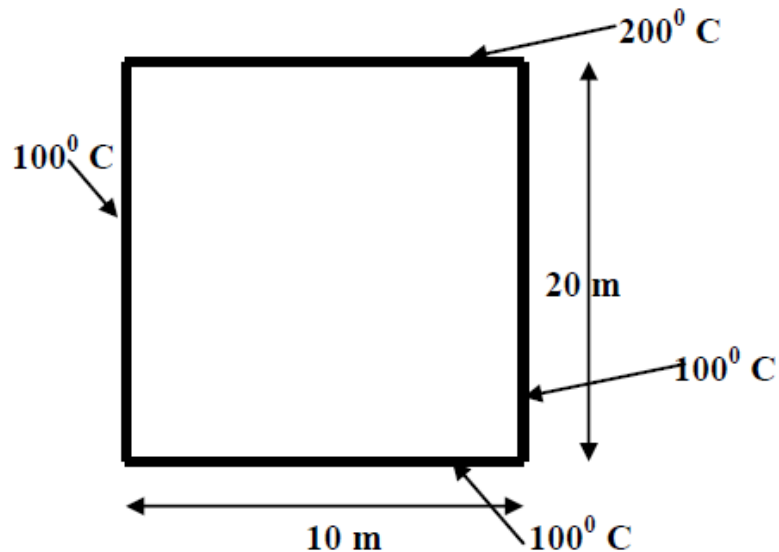


Figure 1

Problem 2. The corner angle bracket is shown below. The upper left hand pin-hole is constrained around its entire circumference and a tapered pressure load is applied to the bottom of lower right hand pin-hole. Compute Maximum displacement, Von-Mises stress.

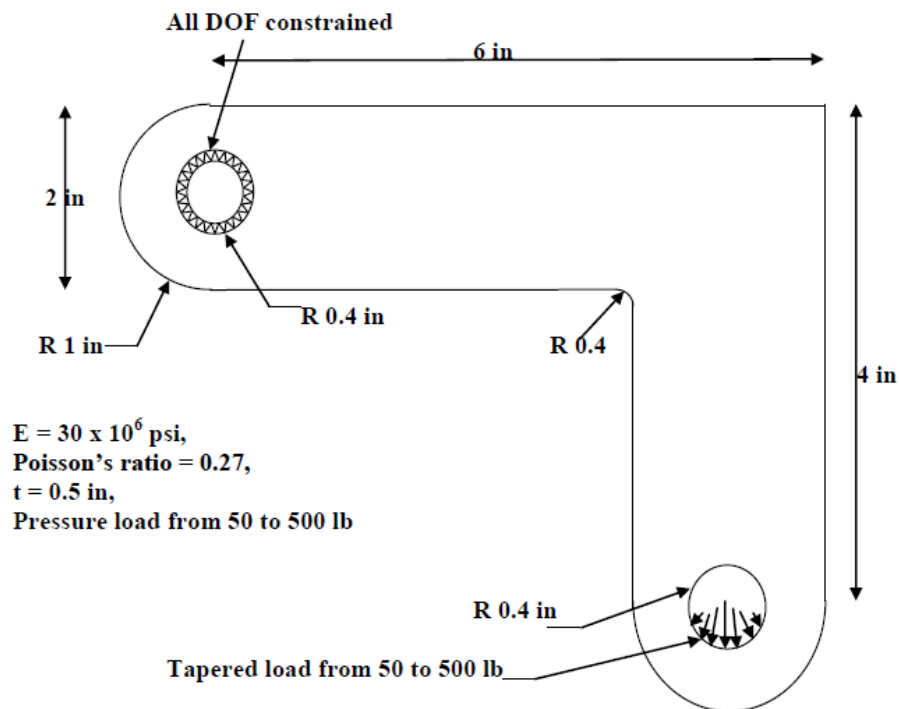


Figure 2

ANSYS Practice Problems

Problem 3. In the Spanner under plane stress, find deformed shape and determine the maximum stress distribution. $E = 2 \times 10^5 \text{ N/mm}^2$, $t = 3 \text{ mm}$, Poisson's ratio = 0.27, Analysis assumption – plane stress with thickness is used.

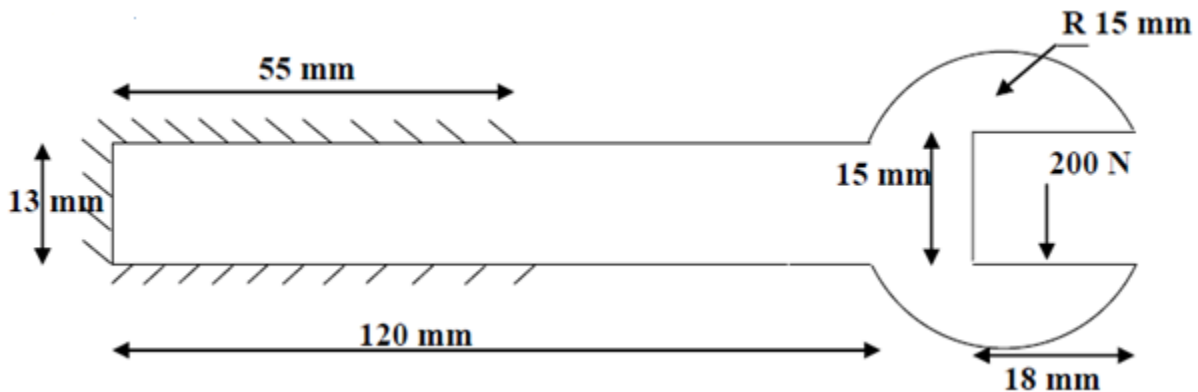


Figure 3

Problem 4. In the plate with a hole under plane stress, find deformed shape of the hole and determine the maximum stress distribution along A-B (you may use $t = 1 \text{ mm}$). $E = 210 \text{ GPa}$, $t = 1 \text{ mm}$, Poisson's ratio = 0.3, Dia of the circle = 10 mm, Analysis assumption – plane stress with thickness is used.

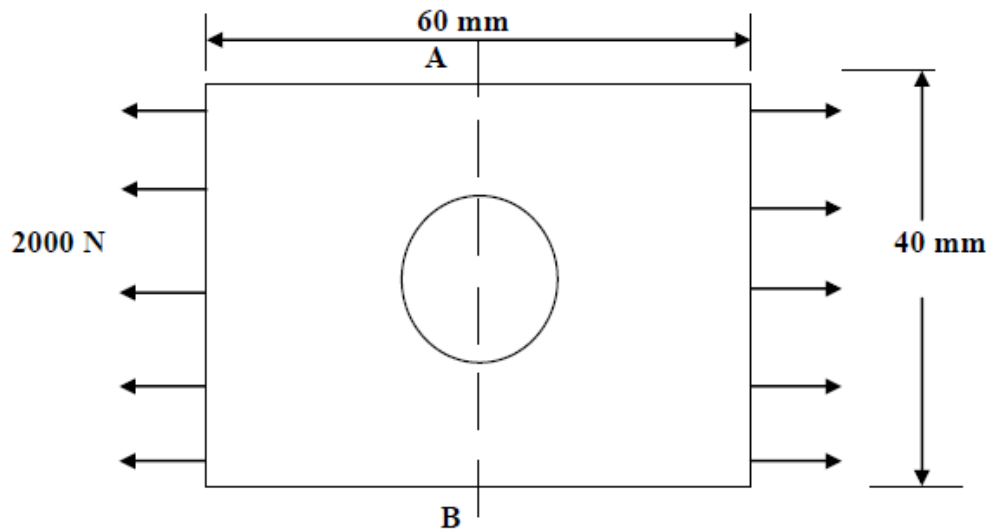


Figure 4

Problem 5. Conduct a harmonic forced response test by applying a cyclic load (harmonic) at the end of the beam. The frequency of the load will be varied from 1 - 100 Hz. Modulus of elasticity = 200 GPa, Poisson's ratio = 0.3, Density = 7800 Kg/m^3 .

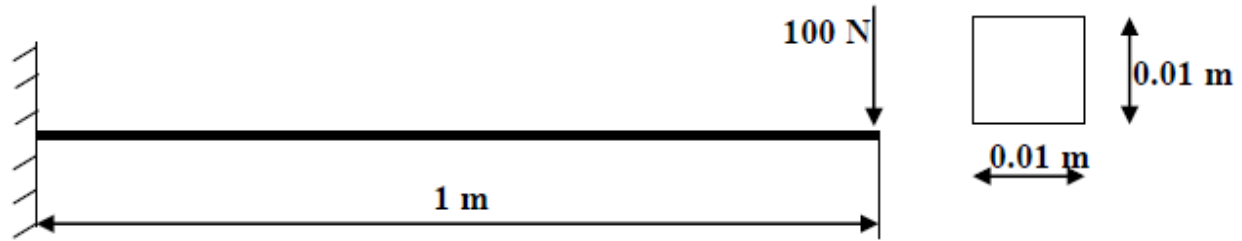


Figure 5

Problem 6. For the given data, find internal stresses developed, Nodal displacement in the planar truss shown in figure when a vertically downward load of 10000 N is applied as shown.

Member	C/s area mm^2	E N/mm^2
1	200	2×10^5
2	200	
3	100	
4	100	

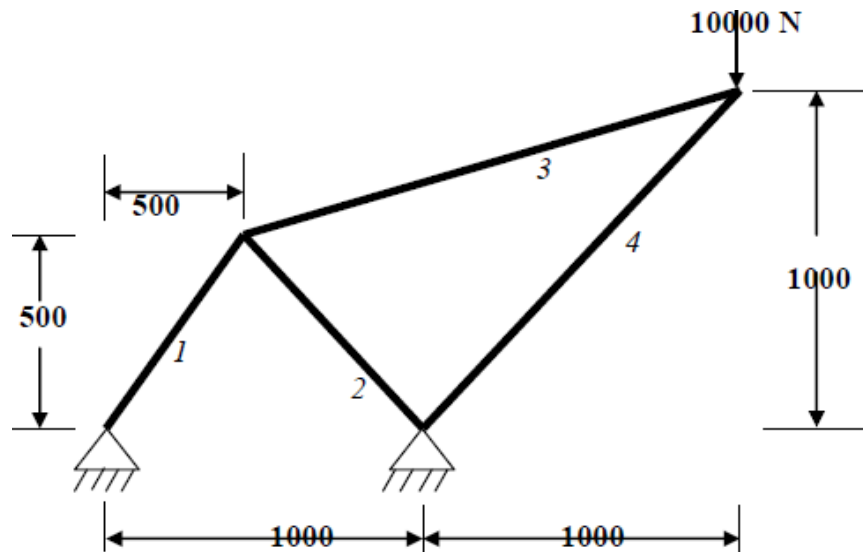


Figure 6

Problem 7. Consider the stepped bar shown in figure below. Determine the Nodal Displacement, Stress in each element, Reaction forces.

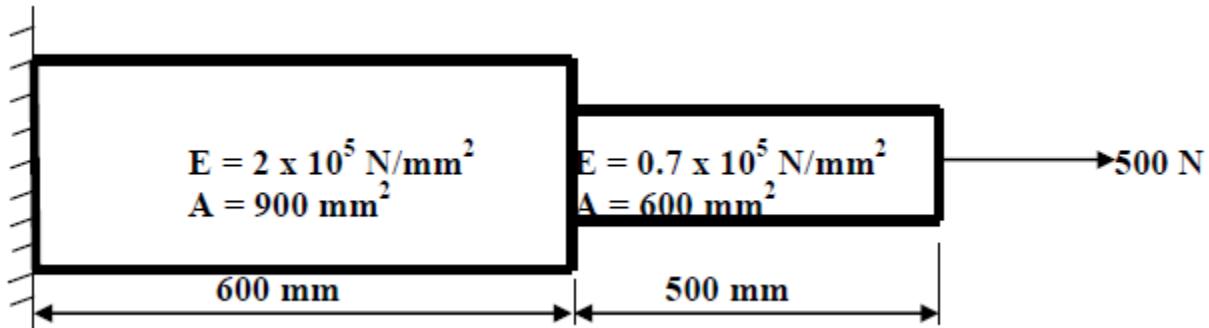


Figure 7

Problem 8. Analysis of a power transmission tower Geometry: the cross section of each of the truss members is $1.56 \times 10^{-3} \text{ sq meter}$. Material: Assume the structure is made of aluminum with $E = 75 \text{ GPa}$.

Boundary conditions: The structure is constrained in the X, Y and Z directions at the bottom three corners. Determine deflection at each joint, stress in each member and reaction forces at the base.

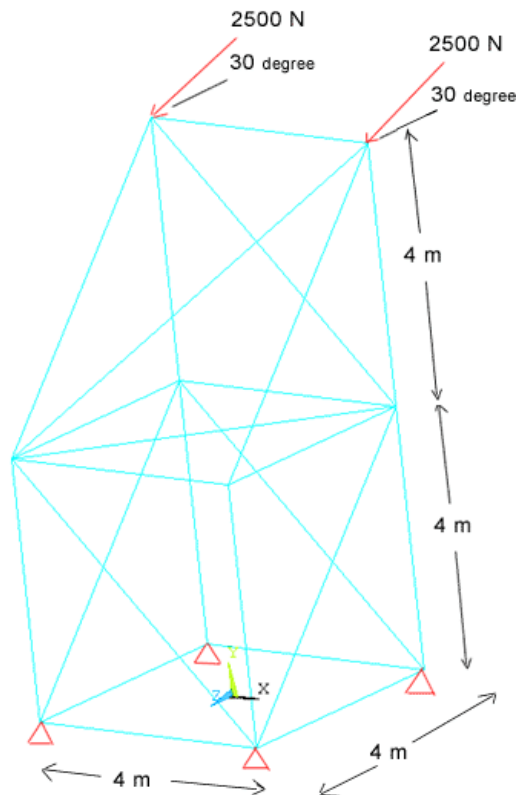


Figure 8

ANSYS Practice Problems

Problem 9. Analysis of a 3D solid object

Material: Assume the structure is made of steel with modulus of elasticity $E=200$ GPa.

Boundary conditions: The object is fixed around the inner surface of the hole.

Loading: The object is loaded uniformly (1000 N/cm^2) along the top surface of the extended beam.

Objective: To plot deformed shape.

To determine the principal stress and the von Mises stress. (Use the stress plots to determine these. Do not print the stress list)

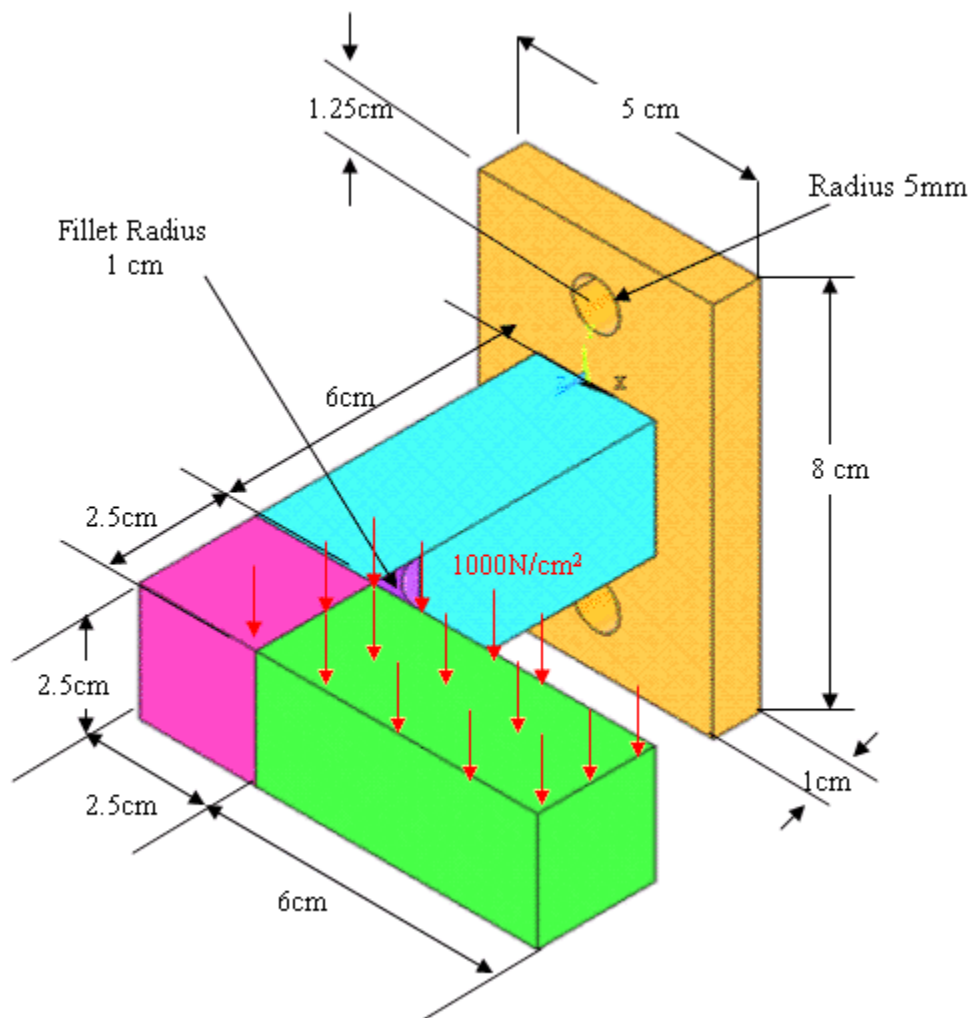


Figure 9

ANSYS Practice Problems

Problem 10. Analysis of a Steel Bracket

The bracket shown below is bolted to a beam at the four bolt holes located in the base. Washers with an outer diameter of 0.7 in are used with the 0.5 in bolts. The bracket is then loaded by a force F of magnitude 2500 lb, at a 45° angle to the horizontal through the hole in the flange. The bracket is made of steel ($E = 30 \text{ Msi}$, $\nu = 0.3$).

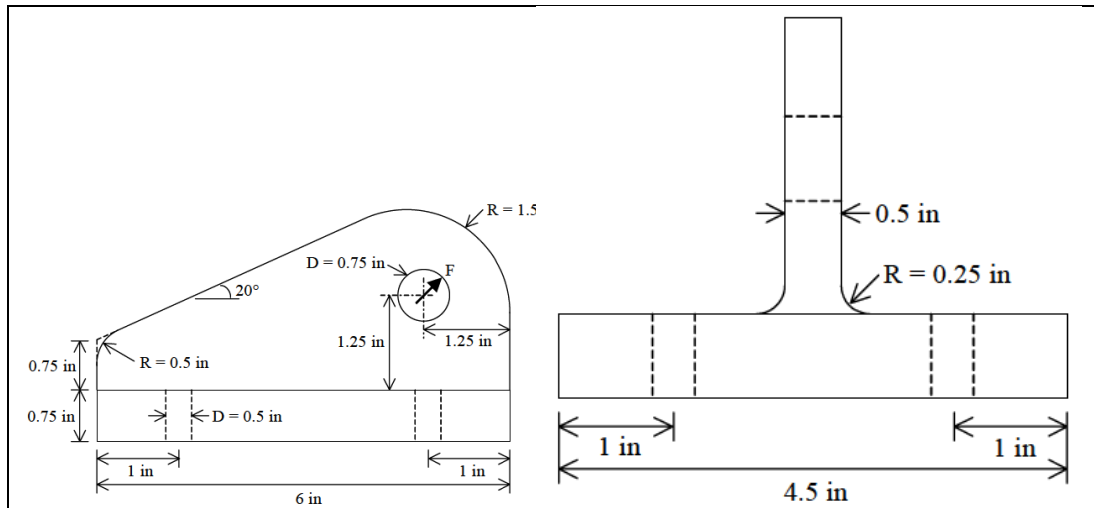


Figure 10

Problem 11. A distributed load of 1000 N/m (1 N/mm) will be applied to a solid steel plate section as shown in the figure below. Model the plate with the central circular hole using standard two-dimensional element and determine the stress around the hole. The dimension of the plate can be considered as 1000mm x 500mm while the modulus of elasticity of the steel is 200GPa. Find appropriate size/ location/ number of the holes required to get near-uniform stress distribution. (Assume the bigger hole diameter is 100mm.)

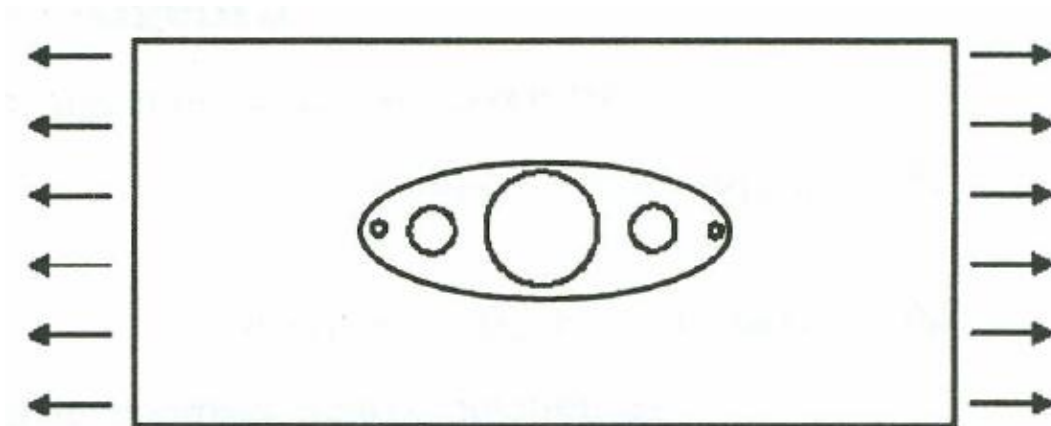


Figure 11

ANSYS Practice Problems

Problem 12. Consider a simplified model of a typical building as shown in figure below. Properties of concrete can be taken as follows: Young's modulus= $1.4 \times 10^{10} \text{ N/m}^2$, Poisson's ratio = 0.3; density= 2240 kg/m^3 . Model the system using beam/plate elements and determine the first few natural frequencies and the corresponding mode shapes.

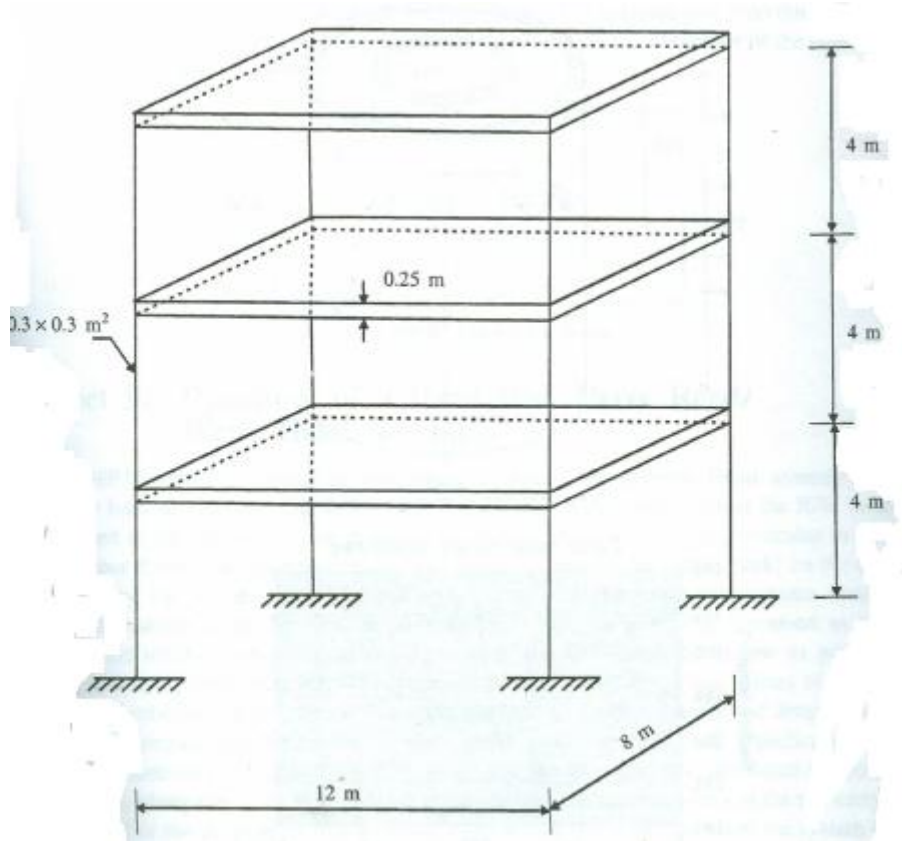


Figure 12